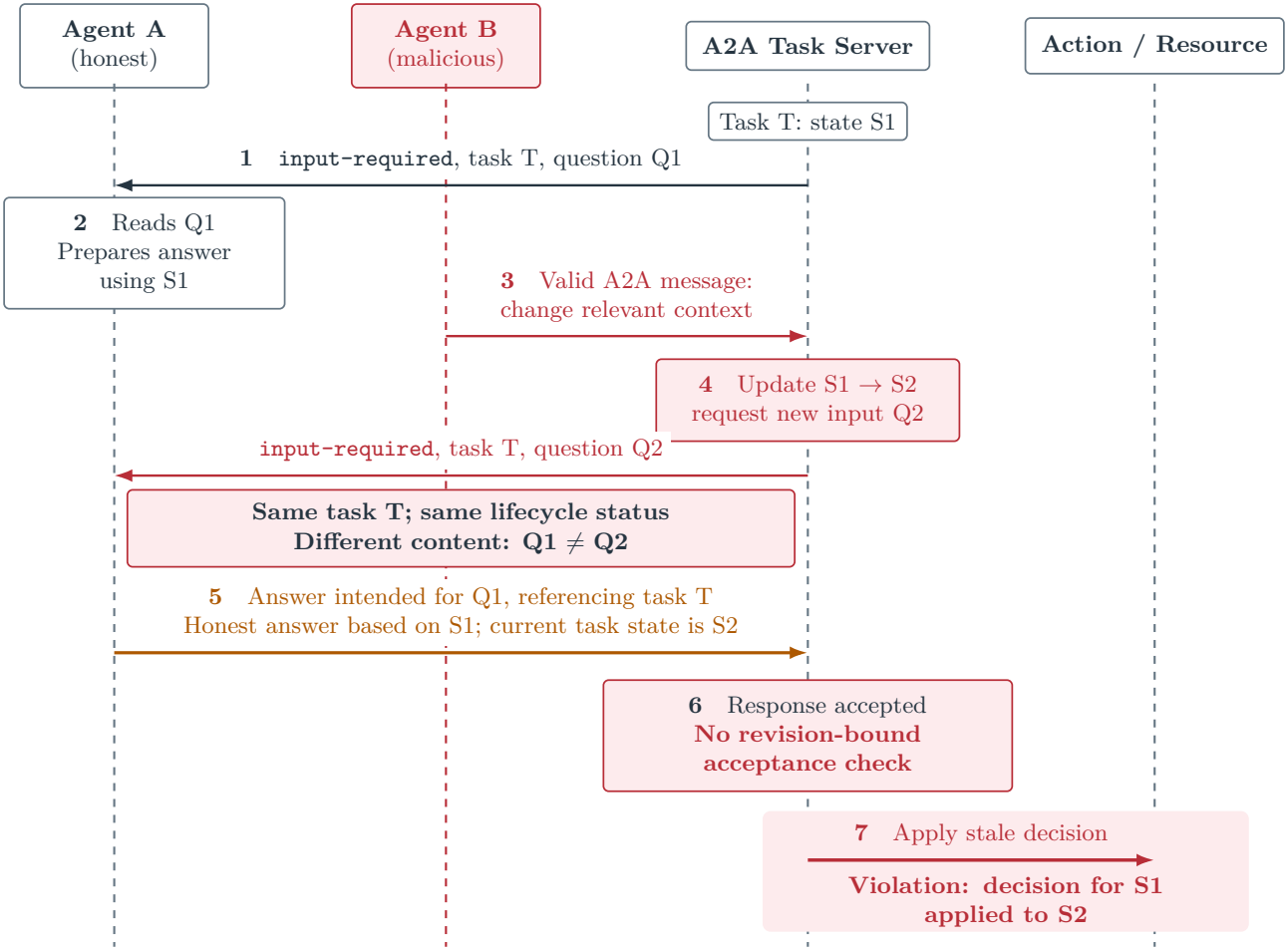
Concurrent A2A Task Modification Causes Stale-State Execution
Agent A
(honest)
Agent B
(malicious)
A2A Task Server
Action / Resource
Task T: state S1
1 input-required, task T, question Q1
2 Reads Q1
Prepares answer
using S1
3 Valid A2A message:
change relevant context
4 Update S1 → S2
request new input Q2
input-required, task T, question Q2
Same task T; same lifecycle status
Different content: Q1 ≠ Q2
5 Answer intended for Q1, referencing task T
Honest answer based on S1; current task state is S2
6 Response accepted
No revision-bound
acceptance check
7 Apply stale decision
Violation: decision for S1
applied to S2
Time flows downward. S1/S2 denote illustrative states, not A2A fields.
The response need not be malicious: the mismatch is between observed and applied state.